

DataForge 2026 | One-Page Concept Summary

DATAFORGE 2026 (PATHWAY TRACK) • NEURIPS 2026 EDUCATION ALIGNMENT

Topic: Synaptic Plasticity as Short-Term Memory	Target Audience: AI Researchers & Data Scientists
Core Subject: Dragon Hatchling (BDH) & BDH-CQ	Falsifiable Claim: Fixed-state synaptic updates eliminate KV cache growth

1. Paradigm Shift: From KV Cache to Synaptic Memory

Standard Transformer architectures preserve sequence context by appending Key-Value vectors to an expanding KV cache. This introduces an $O(N)$ memory footprint per sequence, creating severe GPU RAM bottlenecks and high latency for long-horizon reasoning. **Dragon Hatchling (BDH)** fundamentally departs from this by reformulating attention as dynamic synaptic plasticity: short-term session memory is stored in a fixed-shape connectivity matrix $S_t \in \mathbb{R}^{d \times d}$ updated via local Hebbian writes.

2. Underlying Mechanism & Mathematical Formalism

At each token step t , given Key (K_t), Value (V_t), and Query (Q_t) projections:

• **Synaptic Write (Hebbian Rule):** $S_t = \lambda S_{t-1} + \eta (V_t K_t^T)$, where $\lambda \in (0,1]$ is decay and η is plasticity rate.

• **Synaptic Read:** $V_{hat}_t = S_{t-1} Q_t$ | **Sparse Activation:** $H_t = ReLU(W_x X_t + W_s V_{hat}_t + b)$

Crucial Insight: BDH enforces non-negative sparse activations (~5% active units). This sparsity is mathematically required to prevent catastrophic cross-talk interference within the bounded matrix S_t , enabling monosemantic synaptic connections to encode concepts.

3. BDH-CQ: Latent Test-Time Adaptation Without Backpropagation

Where Test-Time Training (TTT) or HRM/TRM rely on optimization (gradient steps on evaluation samples), **BDH-CQ** acquires unseen task rules in-context without parameter updates or verbalized Chain-of-Thought (CoT) tokens. By carrying forward an evolving recurrent state across demonstration pairs, BDH-CQ shifts the cost-accuracy Pareto frontier—achieving high accuracy on ARC-AGI puzzles with zero inference-time backprop overhead.

4. Architectural Comparison Landscape

Dimension	Standard Transformer	State-Space (Mamba)	BDH / BDH-CQ
Memory Footprint	$O(N)$ Expanding KV Cache	$O(1)$ Recurrent State	$O(1)$ Fixed Synaptic Matrix S_t
Update Mechanism	Concatenate Key-Value vectors	Linear State Transition (A,B,C)	Hebbian Synaptic Write (Outer Product)
Activation Sparsity	Dense (Softmax over N tokens)	Dense continuous state	Sparse ReLU (~5% Active Units)
Test-Time Scaling	Generates CoT Tokens	Recurrent context window	Latent Recurrent State Computation

5. Demonstrated Advantages & Bounded Limitations

- **Key Advantage:** Eliminates KV cache memory explosion; achieves constant-RAM inference over long sequences; enables continuous latent reasoning without CoT token cost.
- **Bounded Limitation:** Fixed matrix capacity S_t implies vulnerability to cross-talk interference when associative key density exceeds the network's sparse capacity channel.

6. Primary References & Sourcing

- Pathway Research (2025/2026). *The Dragon Hatchling Architecture & Equations of Reasoning*.
- Pathway Technical Report (2026). *BDH-CQ: In-Context Learning from Demonstrations without Chain-of-Thought*.